

We have a document D over some alphabet Σ .
Let $\Sigma = \{A, B, C, D\}$

$X = \{ \text{A B C D D A A B C D A A C C D A B C } \dots \}$

Fixed length
binary encoding

$A = 00 \quad B = 01 \quad C = 10 \quad D = 11$

Using this encoding, we can convert X into a seq of 0's & 1's, say Y .

Size of $Y =$ twice the size of X .

$\Rightarrow 200$

We know that no of characters in X is 100
of $A = 60$, # of $B = 25$, # of $C = 10$, # of $D = 5$

Variable length
binary encoding

$A = 0, B = 01, C = 10, D = 1$

$$\text{Size of } Y = 60 \times 1 + 25 \times 2 + 10 \times 2 + 5 \times 1 = 135$$

Avg size $Y = 1.35$

$$\text{Avg encoding length} = \underbrace{60}_{\text{freq of A in X}} \times 1 + \underbrace{25}_{\text{freq B}} \times 2 + \underbrace{10}_{\text{freq of C}} \times 2 + \underbrace{5}_{\text{freq of D}} \times 1 = 1.35$$

Ambiguity: $X \longrightarrow Y$
 AAD/AB 001
 $\times \quad \times$

$AAD \longrightarrow 0\hat{1}10\hat{1}1$

$01 \longrightarrow B$

$\quad \quad \quad \longrightarrow AD$

$\underline{A=0}, D=1$
 $\underline{\underline{B=01}}$

The ambiguity is because of the fact that the encoding of A is appearing at the beginning of encoding of B.

$\equiv$ Encoding of A is appearing as a prefix in the encoding of B.

Prefix free variable length binary encoding:

$A=0, B=10, C=110, D=111$

Avg encoding length = $.6 \times 1 + .25 \times 2 + .1 \times 3 + .05 \times 3$
 $= 1.55$

Problem Statement:

Input: An alphabet Σ and a non-re freq p_a for each symbol $a \in \Sigma$

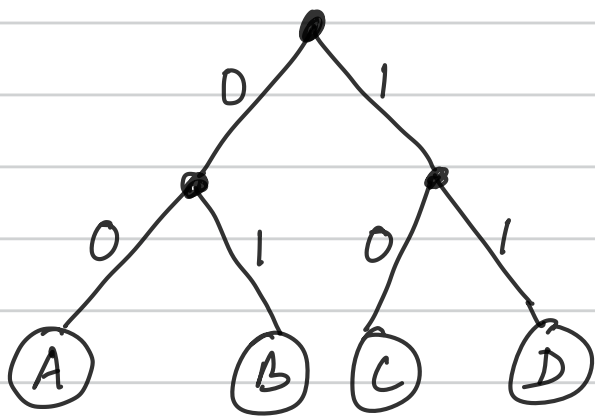
Output: A prefix-free binary encoding of the symbols

in Σ s.t. the avg encoding length is minimized.

min value of $\sum_{a \in \Sigma} p_a \times (\text{no of bits used to encode } a)$

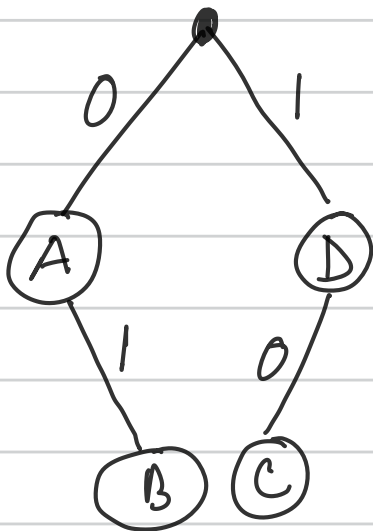
fixed length binary encoding (Prefix free)

$A = 00 \quad B = 01 \quad C = 10 \quad D = 11$



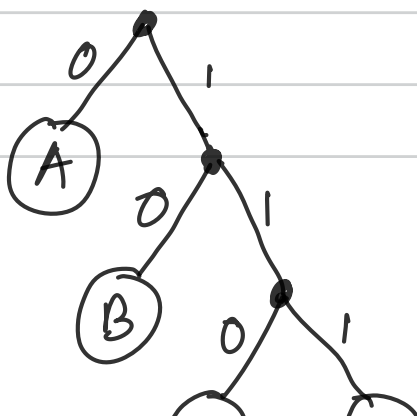
Variable length binary encoding (Not Prefix free)

$A = 0, B = 01, C = 10, D = 1.$



Prefix free binary encoding

$A = 0, B = 10, C = 110, D = 111$



(C) (D)

Prefix free $\Rightarrow$ The leaves of the tree has a one-to-one correspondence with the symbols of Σ . This type of tree is called Σ -tree

Avg encoding length $\equiv$ Avg leaf depth.

For a Σ -tree T & symbol freq $p = \{p_a \mid a \in \Sigma\}$, we denote $L(T, p)$ is avg depth of a leaf in T .

$$= \sum_{a \in \Sigma} p_a \times (\text{depth of the leaf labeled as 'a' in } T)$$

Problem Statement

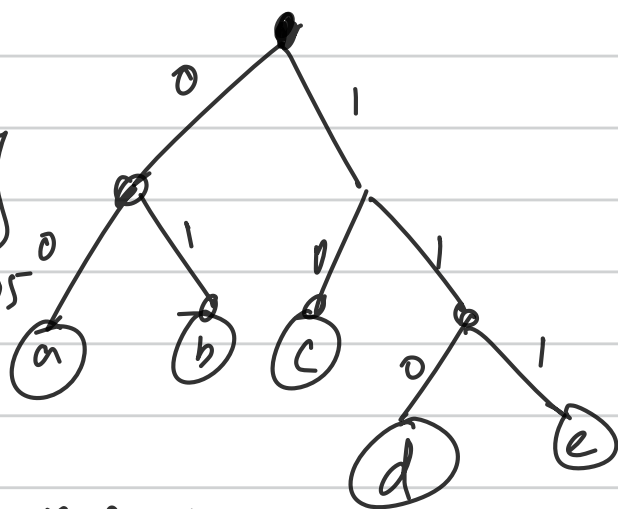
Input: Same

Output: A Σ -tree with min avg leaf depth.

$$\Sigma = \{a, b, c, d, e\}$$

$$p = \{p_a, p_b, p_c, p_d, p_e\}$$

$\cdot 3 \quad \cdot 3 \quad \cdot 3 \quad \cdot 0.5 \quad \cdot 0.5$



Avg leaf depth =

$$\cdot 3 \times 2 + \cdot 3 \times 2 + \cdot 3 \times 2 + \cdot 0.5 \times 3 + \cdot 0.5 \times 3$$

$$= 2.1$$

$$a = 00$$

$$b = 01$$

$$c = 10$$

$$d = 110$$

$$e = 111$$

$p_d, p_e \rightarrow$ should be smaller

Forest

(A)

.6

(B)

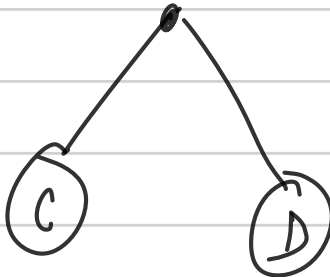
.25

(C)

.1

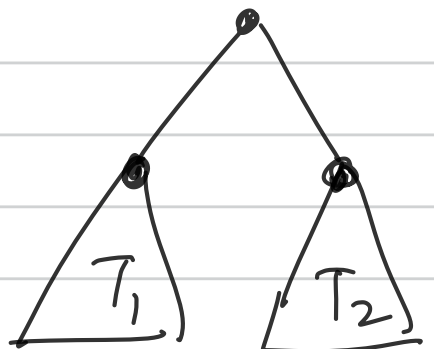
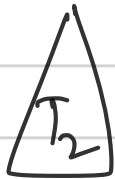
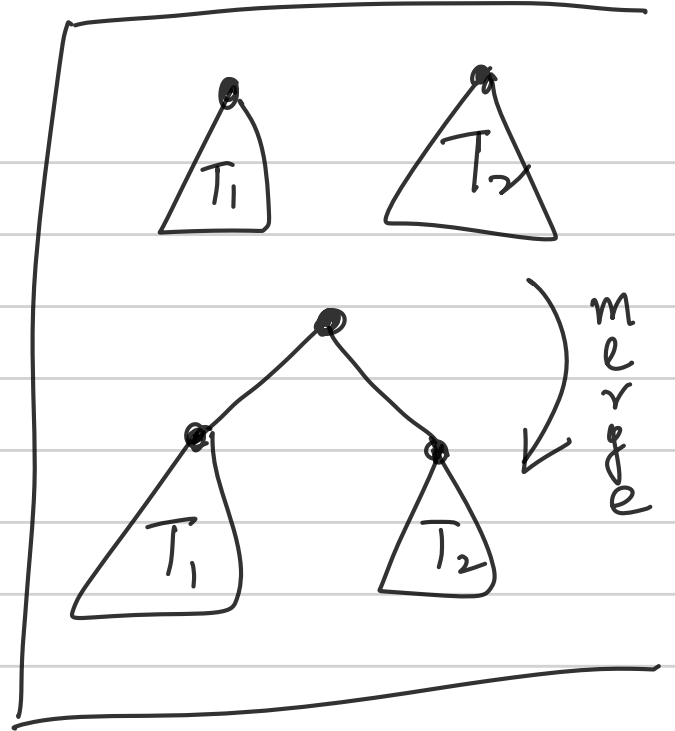
(D)

.05



$$.1 \times 1 + .05 \times 1$$

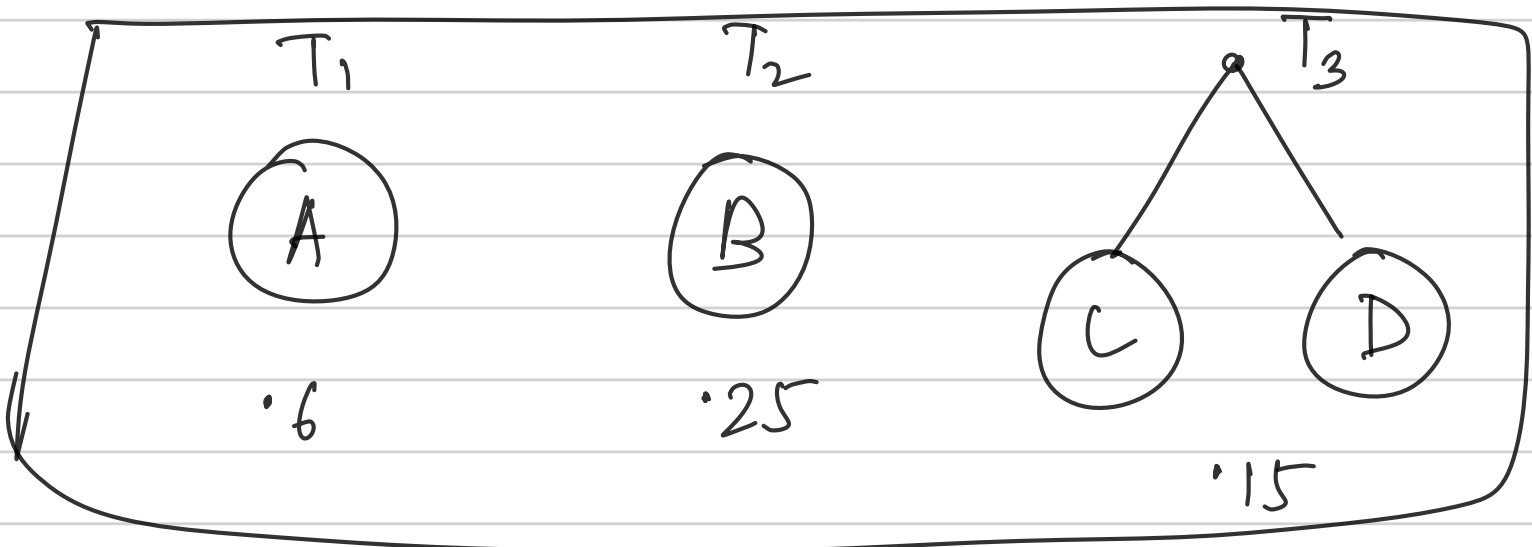
$$= .15$$

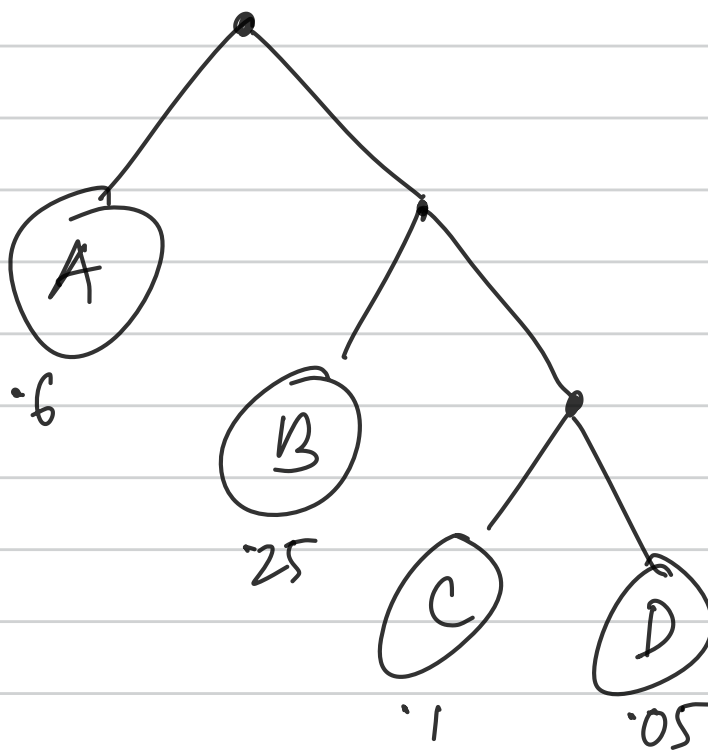
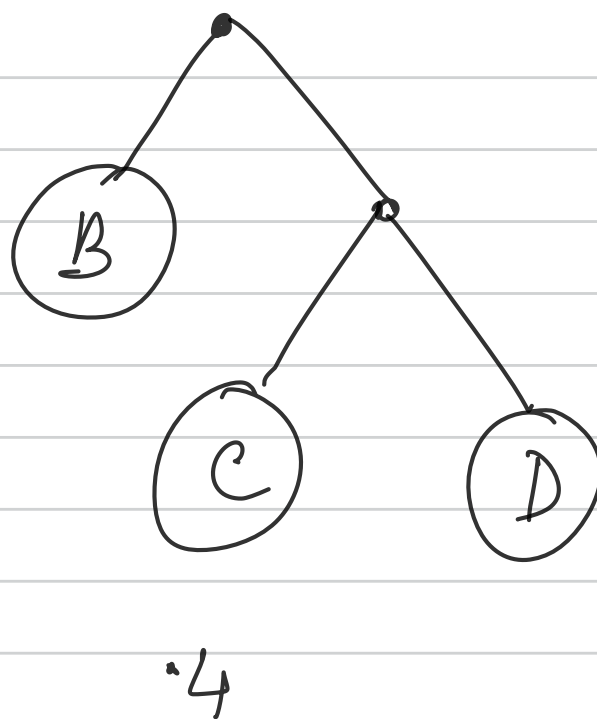


$$\text{increase in avg leaf depth.} = \sum_{a \in T_1} p_a + \sum_{a \in T_2} p_a$$

We decide to merge two trees T_1 & T_2 s.t.

$$\sum_{a \in T_1} p_a + \sum_{a \in T_2} p_a \text{ is } \underline{\underline{\text{min}}}.$$





Pseudo Code

for each $a \in \Sigma$

$T_a = \text{tree containing one node labeled as } (a)$

$P(T_a) = p_a$

$\mathcal{F} = \{T_a / a \in \Sigma\}$

While $\mathcal{F}$ contains atleast 2 trees.

$\{ T_1 = \text{tree with min } P(T_1) \text{ in } \mathcal{F}$

$T_2 = \text{tree with 2nd min } P(T_2) \text{ in } \mathcal{F}.$

$T_3 = \text{merge}(T_1, T_2) \quad P(T_3) = P(T_1) + P(T_2)$

$\mathcal{F} = (\mathcal{F} \setminus \{T_1, T_2\}) \cup T_3$
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Return the unique tree in $\mathcal{F}$.